High Order Field-to-Field Corrections for Imaging and Overlay to Achieve Sub 20-nm Lithography Requirements

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ABSTRACT

Immersion lithography is being extended to the 20-nm and 14-nm node and the lithography performance requirements need to be tightened further to enable this shrink. In this paper we present an integral method to enable high-order field-to-field corrections for both imaging and overlay, and we show that this method improves the performance with 20% - 50%. The lithography architecture we build for these higher order corrections connects the dynamic scanner actuators with the angle resolved scatterometer via a separate application server. Improvements of CD uniformity are based on enabling the use of freeform intra-field dose actuator and field-to-field control of focus. The feedback control loop uses CD and focus targets placed on the production mask. For the overlay metrology we use small in-die diffraction based overlay targets. Improvements of overlay are based on using the high order intra-field correction actuators on a field-to-field basis. We use this to reduce the machine matching error, extending the heating control and extending the correction capability for process induced errors.

Keywords: 20-nm, 14-nm, Immersion lithography, CDU, Overlay, Focus, diffraction based overlay, holistic lithography,

1. INTRODUCTION

In 1973 the very first projection lithography tool was introduced to the market. This tool, the Perkin Elmer Micralign, was used to print features of about 3-µm devices with overlay accuracy close to 1-µm. 40 years later, optical projection lithography is more alive than ever, and it is being used to pattern 2X-nm DRAM devices, 20-nm logic devices and 1Y-nm NAND devices, with overlay accuracy close to 5-nm. As such, we can conclude that over 40 years projection lithography enabled an effective area shrink of about 10^4 times while the lithography accuracy improved about three orders of magnitude. Using overlay as an example, figure 1 illustrates the performance improvements over time.

Over these four decades we can recognize different eras of innovation. During the 1980ties the understanding of overlay has grown, and linear optimization models linked to projection tool capabilities have been derived and used for system setup improvements. In the 1990ties process control in the wafer fab has been adopted widely and metrology feedback has been a main driver to reduce overlay. Later, with the introduction of more complex scanner systems and advanced projection lens manipulators, higher order wafer corrections have become available. Starting from about 2007 a holistic approach has been adopted to handle this novel complexity. In the holistic approach the lithography performance is optimized in both design phase as well in manufacturing phase. In the design phase the process windows and metrology target robustness for process variations is maximized, whereas in the manufacturing phase the performance variation is minimized using dynamic process window control. Latter is the main topic of this paper.

In this paper we will discuss the systems architecture we have been building to support holistic lithography optimization for imaging and overlay. Specifically we discuss scanner setup improvements for matched-machine overlay, reticle heating control for improved on-product intra-field overlay and the usage of high-order field-to-field corrections for imaging, focus and overlay. These corrections are needed to bring on-product overlay performance to 4-nm and CD uniformity to 1-nm. These lithography performance levels are supporting immersion lithography down to the 14-nm node. A more detailed description of the systems architecture we build to enable these corrections is given in section 2.

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In section 3 we will discuss the experimental results demonstrating the predicted imaging and focus performance improvements. The overlay experimental results and resulting overlay performance roadmap are discussed in section 4. Finally in section 5 we summarize and give our conclusions.

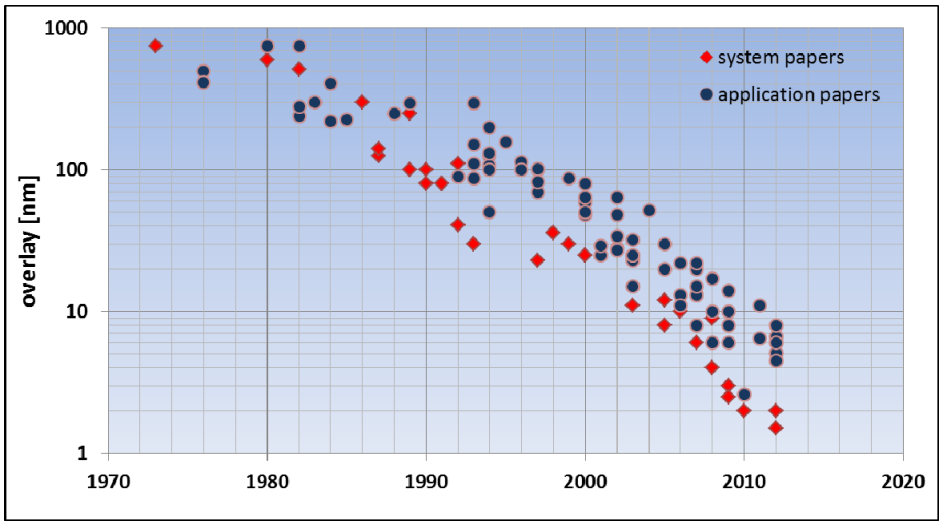

Figure 1: An overview of published overlay results between 1973 and 2013 [1].

2. ARCHITECTURE FOR LITHOGRAPHY PERFORMANCE CONTROL

The architecture for process window control is shown in figure 2. We can recognize three main building blocks: 1) scanner actuators interface; 2) optical scatterometry for overlay, focus and CD metrology; and 3) an application server for sampling optimization and model based control.

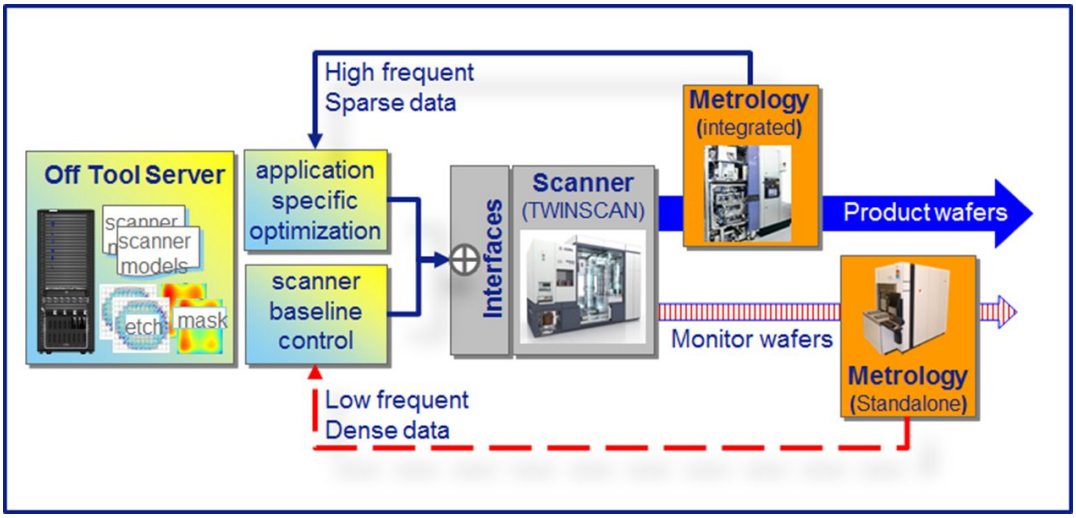

Figure 2: Lithography optimization systems architecture for process window control

The scanner actuator interface is used for applying recipe information to the illumination system, projection lens and scanning stages. These user defined set-points of the actuators are used for scanner generic and application specific performance control. The scanner generic optimization is used for machine to machine matching and long term machine stability control. The method applied here is based on exposing monitor wafers with reticles that contain many test targets, such that with high density sampling the intra-field and inter-field fingerprints for overlay, focus and CD uniformity can be determined. The monitor wafer flow is running in parallel with the production wafer flow, but the frequency of control updates is lower: typically 1 or 2 times a week. On the other hand, the updates of the settings for the production wafers, is high frequent (once per lot) and is based on a more sparse sampling of the imaging and overlay fingerprints. The scanner actuator interface is embedded in the internal scanner lot control system. The lot control uses the on-board sensors for dose, imaging, alignment and focus to minimize thermal drifts. By embedding the external scanner actuator interface in the internal scanner lot control it is possible to maximize the available actuator ranges for user defined offsets while keeping the system lot control full functional.

The second building block of the lithography performance control system is an angle resolved scatterometer [2,3,4]. ASML has developed this high NA angle resolved scatterometer for scanner performance control. The scatterometer combines measurement of overlay, focus and CD in one sensor. In an angle resolved scatterometer the metrology signal is observed in the pupil of the metrology objective. The targets for overlay and focus are designed such that the signal of the first order diffraction is measured. Because of the small resolution and the relatively high detection wavelength (>425nm), the pupil for CD metrology collects only the zero order, which is than rigorously reconstructed to retrieve the CD profile of the measured pattern.

The metrology tool is available as a standalone version and a wafer track integrated version. Latter is of importance since it enables high frequent and fast updates of the scanner recipe for imaging and overlay corrections, whereas the standalone metrology tool is used for recipe creation and long term stability control based on the monitor wafers. The scanner corrections for production wafers are based on the use of special targets for overlay, focus and CD placed on the production mask. For overlay measurements a diffraction based overlay (DBO) method with small (10um x 10um) in-die targets (micro DBO) is developed. The in-die targets are important since they enable the metrology of the high order terms of the intra-field fingerprint. For focus we developed a similar diffraction based focus (DBF) method looking to the asymmetric side wall angle printed with these targets. This method is described in separate papers [5,6]. The asymmetric focus targets can measure focus offsets and focus tilts per exposed field. The CD metrology concept of the angle resolved scatterometer uses the target reflection spectrum as a function of the angles of incidence in the pupil of the high NA sensor lens. The simulated reflection spectra are computed from a model (a fast Maxwell solver) and by comparing the measurement with the model parameters CD and SWA describing the details of the line profile are determined. The across field and across wafer CD fingerprints are either measured on the device (in case of repetitive design) or measured on targets mimicking the device.

The third building block of the performance control architecture is an off-tool lithography computation platform (LCP) server. The LCP is designed to serve multiple scanner and metrology tools at the same time. The server is used to compute the optimum lithography recipe for the each scanner and each application. This LCP software contains computational models of the scanner, actual scanner performance data and fingerprints and calibration data for the application specific optimizations. For the integrated metrology system, the LCP is used to calculate the optimum sample schemes to enable dynamic high-order scanner corrections.

3. IMAGING PERFORMANCE OPTIMIZATION

3.1 CD uniformity and focus flow

CD uniformity control makes use of the active dose and focus control actuators in the scanner system. The light pulses created in the excimer laser source are controlled in dose for every individual pulse, thus creating intra-field dose profiles in the scan direction of the exposed field. Across slit dose is actuated with the illuminator system, and together with the

dose per pulse control of the laser a vector based freeshape xy dose uniformity pattern can be created. This can be done for every individual field. Focus control makes use of the stage z-positioning in 6 degrees of freedom. Per exposed field the offsets and tilts in X and Y can be set.

The optimization of CD uniformity and focus is described in the scheme presented in figure 3. In a first step the DBF and CD targets must be designed. For DBF the target design is such that for the given scanner illumination setting and the given wafer process stack the printability of the target is optimized and the scatterometer signal is maximized for detectability of focus signals and robustness against other process perturbations such as dose. Depending on the needed process corrections, the target density on the production mask is determined and designed, this for both DBF targets as well as CD metrology sites. In a second step the sensitivity of the targets for focus and dose is determined. This is done by exposing a focus-expose-matrix (FEM) of the product mask on the wafer. The DBF targets and CD targets are measured as function of focus and dose to create the calibration curves. Now that the optimization cycle has been initialized, the product wafers can be exposed, and during the production process the scatteromer measures the inter-field and intra-field variance of focus and CD and calculates the correctables for the subsequent lots. Finally, the corrections are applied.

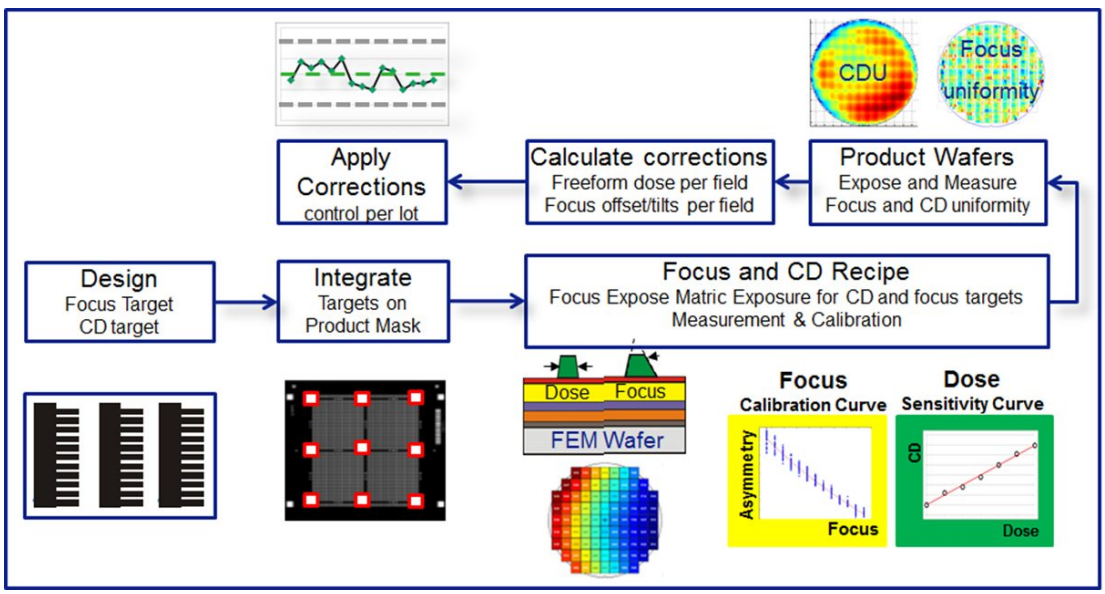

Figure 3: Optimization flow for CD and focus uniformity from target design to control on product wafers.

3.2 Experimental Results

For a DRAM use case we investigated the benefit of this imaging performance optimization. First, in figure 4 we show the experimental results for CD uniformity. In this case the main interest of the user was the optimization of the intra-field CD fingerprint and for this a highly dense sample layout of 17x25 CD metrology points (425 points) was used. Per wafer 5 fields where averaged to obtain the reported CD uniformity number. The CD metrology was based on the full CD and wafer process stack reconstruction. The correction of the CD was done for each of the 425 points individually at and applied at the same time. In this example use case we find that CD uniformity can be improved towards 0.4-nm, which is 44% better when compared to no-correction case, or 17% better when compared to already available 6th order intra-field dose corrections.

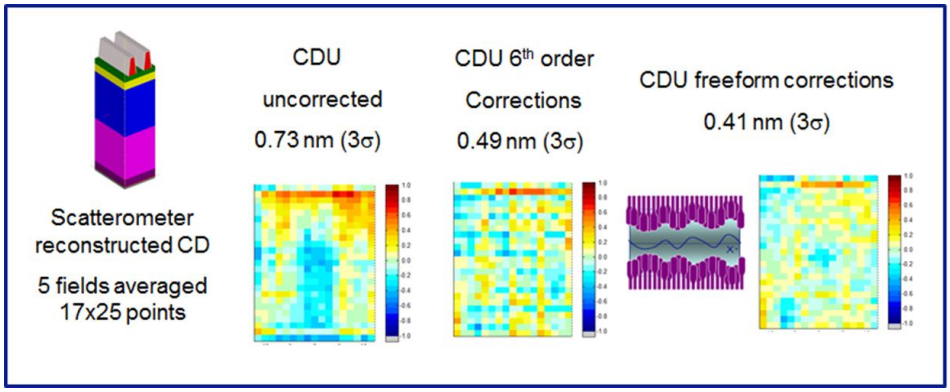

Figure 4: Results of optimization of intra-field CD uniformity fingerprint using free-form dose corrections.

For across wafer focus optimization on production wafers, we developed a new method based on diffraction based asymmetric focus targets. For scribe line usage we use a DBF target with size of 40umx40um, and later for in-die usage we will use uDBF targets with size of 10umx10um. We expect that the typical application will use few marks on the reticle and corrects focus offsets per field. For ultimate performance the Rx, Ry focus tilts corrections will be used with more uDBF targets placed per field. To investigate the benefit of on-product focus optimization we investigated a logic use case, the experimental results of which presented in figure 5. The test was done on a full wafer exposure with sample layout of 4x8 per field. The test was executed on a monitor wafer (resist on silicon) and on a back-end-of-line (BEOL) process stack wafer. The applied field-to-field corrections include focus offset, and focus tilts Rx and Ry. The result shows that focus uniformity is improved towards 18-nm, which is an improvement of about 25 % to 35 %. This is very significant, especially since the process latitude for focus for these BEOL layers is small.

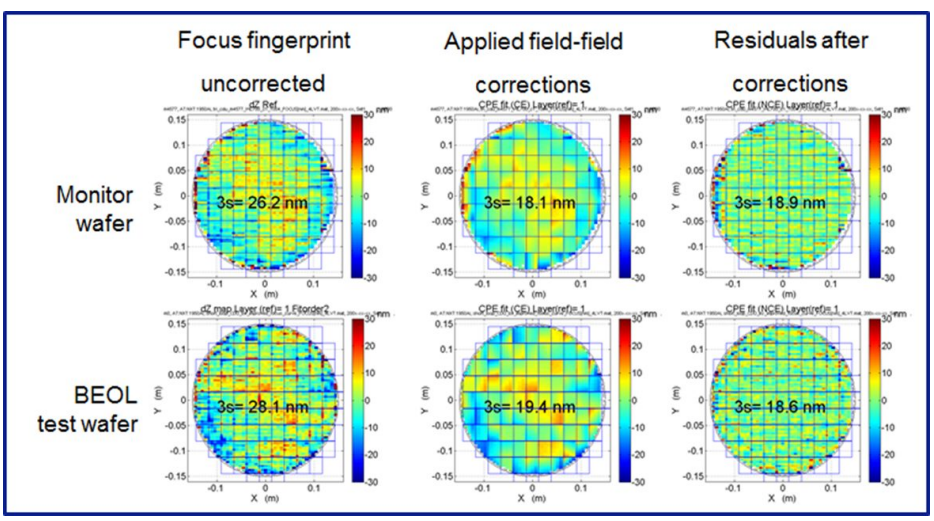

Figure 5: Results of optimization of across wafer focus uniformity using diffraction based focus targets.

More experimental results on both CD and focus optimization using the angle resolved scatterometer and the high order field-to-field corrections are presented in a separate paper [6].

4. OVERLAY PERFORMANCE OPTIMIZATION

4.1 Reducing the overlay error budget with high-order field-to-field corrections

In figure 6 we show the on-product overlay error budget breakdown and the overlay improvement roadmap for immersion lithography. This is based on a contact to gate use case, starting with 32-nm node and moving to the 14-nm node. The budget assumes full wafer sampling and full lot metrology evaluation and is given for a matched machine usage and a dedicated chuck usage. In matched mode usage it is assumed that the different layers are randomly exposed on different machines, and in de dedicated mode it is assumed that the logistics of the wafers is organized such that all critical layers are exposed on one machine with dedicated chuck.

The on-product overlay performance is determined by the scanner and the scanner's interaction with the different process layers. The latter part we call application specific part of the budget, whereas the first part is system generic. As can be seen in the budget, the intrinsic systems overlay performance is only a small part of the budget. This part is easily measured and is part of the systems acceptance testing, just as the matched machine overlay. The application parts of the budget are depending on the used reticles, used resist process and the layers on the wafer. Depending on previous process steps the wafers can be deformed and the alignment and overlay targets printed on the wafer can be deformed. The reticle content (effective transmission) and the required resist dose determine the lens and reticle heating effects, and with dedicated testing these effects can be quantified. The wafer deformation and mark deformation is more difficult to quantify from simple testing. A more in depth study is needed per process step to quantify these contributors. Metrology and APC errors are part of the budget, but are relatively small.

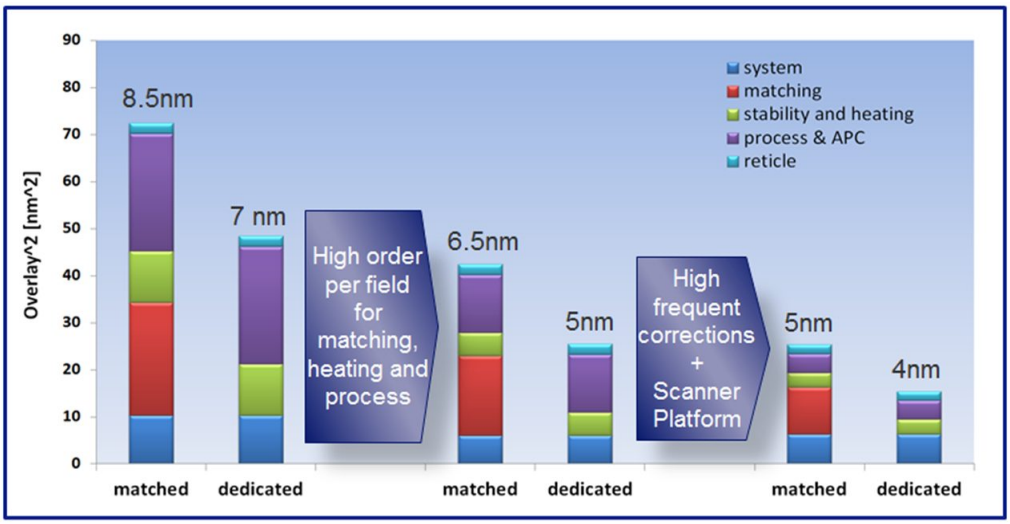

Figure 6: On-product overlay roadmap for immersion lithography for contact to gate use case assuming full wafer sampling and full lot.

Next to intrinsic scanner platform improvements, the roadmap is based on reducing the error components for matching, heating and process. These three components can be reduced by applying higher order corrections for each individual exposed field (i,j). In equation (1) and in table 1 we present the polynomial coefficient definition up to third order intra-field. The correction vector (dx, dy) as defined by this polynomial is already available as lot average intra-field process correction (iHOPC). New for the latest scanners is that with new lens control this function becomes available per field, and this without impact on the productivity of the scanner.

$$\begin{pmatrix} dx \\ dy \end{pmatrix}_{i,j} = \begin{pmatrix} k_1 + k_3 * x + k_5 * y + k_7 * x^2 + k_9 * xy + k_{11} * y^2 + k_{13} * x^3 + k_{15} * x^2 y + k_{17} * xy^2 + k_{19} * y^3 \\ k_2 + k_4 * y + k_6 * x + k_8 * y^2 + k_{10} * yx + k_{12} * x^2 + k_{14} * y^3 + k_{16} * y^2 x + k_{18} * yx^2 + k_{20} * x^3 \end{pmatrix}_{i,j} \quad (1)$$

Depending on scanner system type and depending on correction algorithm a subset of these 20 parameters is available for field-to-field adjustments. In the next sections it will be shown how these actuators improve scanner matching, scanner heating control and correct for process overlay effects.

Table 1: High order correction parameters for field-to-field corrections

Parameter	coefficient	Term	Actuator	Parameter	coefficient	term	actuator
Translation	K1		Scan	Bow	K11 (X)	Y2	Scan
	K2		Scan		K12 (Y) (D2y)	X2	Lens
Magnification	K3 (Mx)	X	Lens	3rd-order magnification	K13 (X) (D3x)	X3	Lens
	K4 (My)	Y	Scan		K14 (Y)	Y3	Scan
Rotation	K5 (Rx)	Y	Scan	Accordion	K15 (X)	X2Y	Scan Lens
	K6 (Ry)	X	Scan		K16 (Y)	Y2X	Scan
2nd order magnification	K7 (X) (D2x)	X2	Lens	C-shape distortion	K17 (X)	XY2	Scan Lens
	K8 (Y)	Y2	Scan		K18 (Y)	YX2	Scan Lens
Trapezoid	K9 (X)	XY	Scan Lens	3rd-order-flow	K19 (X)	Y3	Scan
	K10 (Y)	YX	Scan		K20 (Y) (D3y)	X3	-

4.2 Experimental results: Scanner and lens matching

As figure 6 already indicated, for matched machine usage, the grid matching component is the largest error in the overlay budget. Machine matching for inter-field as well as intra-field is done using the baseline monitor wafer flow as described in section 2 and figure 2. In order to improve the grid calibration we developed a new method based on multi-shift exposures to pick up spatial grid errors. In figure 7 the basics of this method are illustrated. In the scanner system the stage positioning is done using a grid plate encoder system. For spatial frequency below ~15 mm the grid plates are first measured at the supplier and then additionally calibrated during scanner setup using sensor data of the stage positioning system; the residual high-frequency errors are acceptably low. For spatial frequency above ~15 mm other system contributors become important as well, and the inter-field wafer grid must be calibrated. Here we expose full fields with overlay and alignment targets on a reference wafer with etched marks on a 13x19 intra-field grid. The reference wafers approximate a perfectly calibrated scanner that has a very straight grid. The wafers have etched alignment marks and DBO targets such that the grid calibration can be done with either the alignment sensor or with the angle resolved scatterometer. By exposing fields with dual and multi-shifts over the 13x19 grid we sample the inter-field grid error. By using the stage actuators we can calibrate the grid errors towards the reference wafer. For intra-field grid calibration we do the same, now using the higher order lens and stage actuators. This method results in a field layout independent grid matching. An additional correction with the higher order process corrections is possible, however, will make the matching calibration layout dependent.

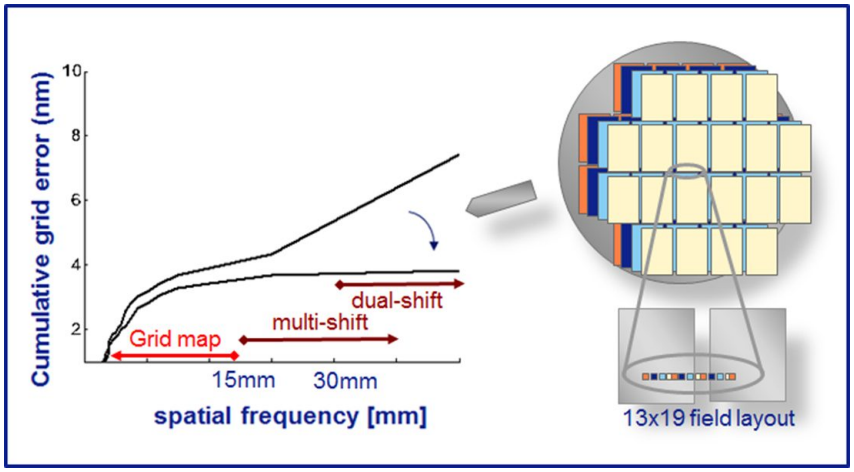

Figure 7: Illustrating the new grid calibration method using multi-shift exposures to sample spatial grid errors

In figures 8 and 9 we present results of this method on the machine-to-machine matching. First figure 8 shows the result of the intra-field distortion fingerprint after calibration: the maximum distortion error is 1.6-nm, and after applying additional 15 parameter corrections this reduces to 1.2-nm.

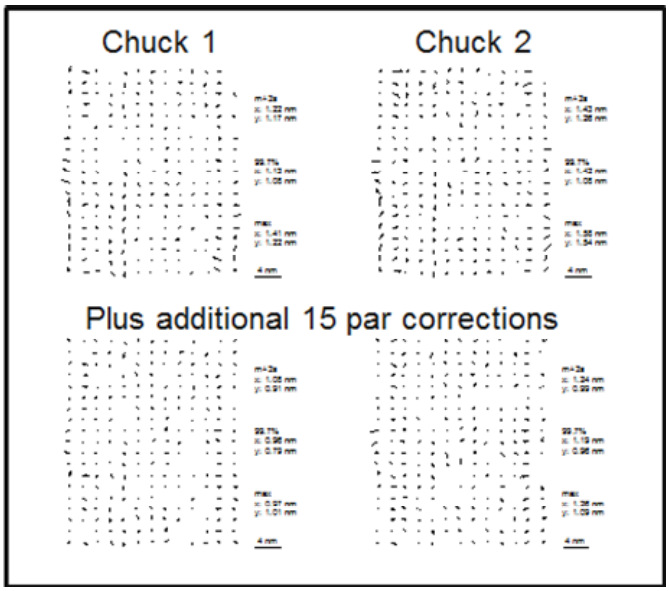

Figure 8: Experimental results of intra-field distortion after calibrating the immersion tool, with on the top row the status after improved grid setup and the bottom row after applying an additional 15 parameter correction.

For across tool matching we performed an experiment involving 5 different immersion scanners. The reference wafers are used to calibrate the 5 different machines. In the left chart of figure 9 we present an example result of the achieved matched overlay with respect to the reference wafer. An overlay of 3.2-nm is obtained (99.7% criterion for full wafer). The other machines have similar performance towards the reference wafer. Looking to the machine-to-machine overlay, we find matched overlay values between 3.0-nm and 3.8-nm (right plot of figure 9).

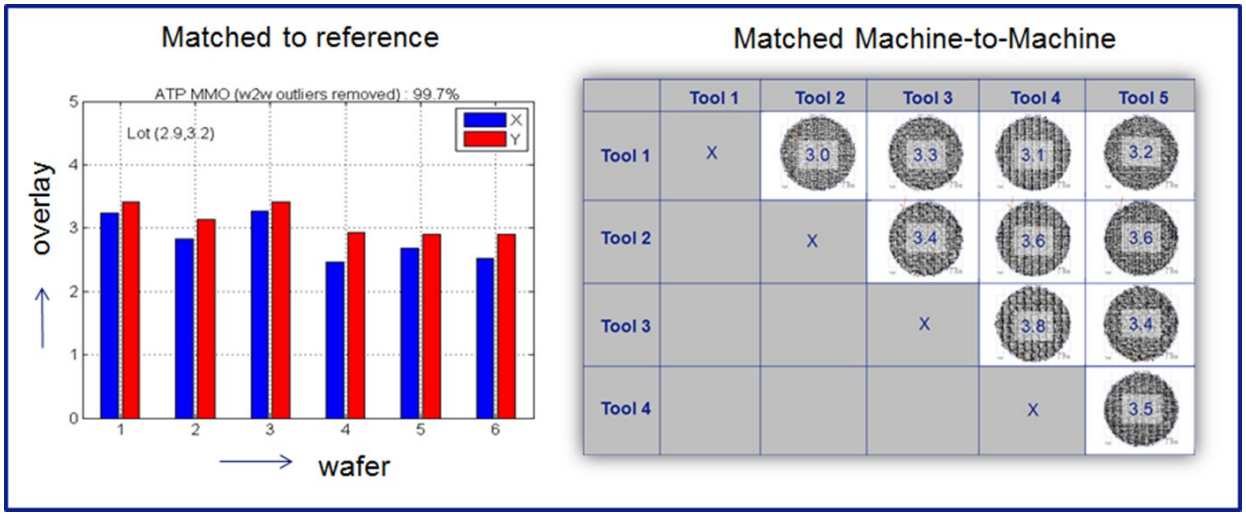

Figure 9: Experimental results of the new machine matching method. Left plot shows the matched overlay of a 6 wafer lot to the reference wafer. The table on the right shows the result of the cross matching a pool of 5 tools.

The matched machine overlay can be improved further with high order corrections; however, this correction is layout dependent. In figure 10 we present the results of such an additional correction, and is the data shows that the matched machine overlay is reduced to 2.5-nm for the lot of wafers exposed. The correction applied was higher order per field as explained in section 4.4. The dynamic distortion was reduced to about 1-nm.

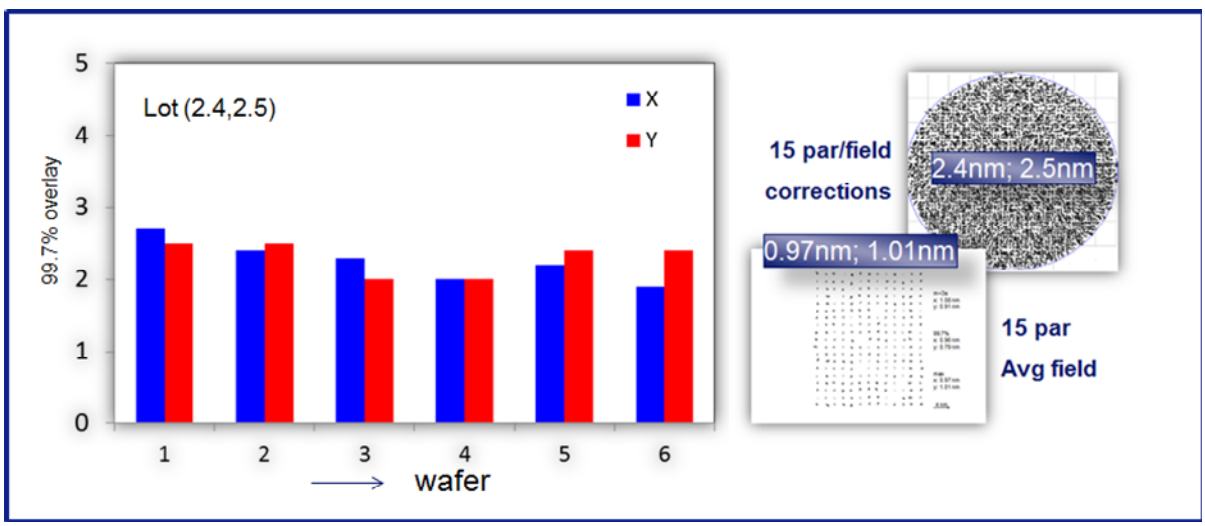

Figure 10: Experimental results of matched machine overlay after applying the new machine grid calibration method and an additional 15 parameter per field correction.

4.3 Experimental results: Reticle heating control

Overlay heating effects come from the projection lens and from the reticle. The projection lens thermal aberration control improvements are discussed in a previous paper [7]. In the same paper we also presented the status of reticle heating control. Reticle heating is caused by light absorption in the absorption layer on the mask, and results in deformation of the mask blank. Reticle heating is especially seen in low transmission mask. The mask deformation is partly corrected by

the wafer to wafer reticle alignment, but with decreasing overlay requirements additional measures are needed. In the latest generation immersion tools we implement a further extension of the reticle heating control, now including 18 parameters field-to-field overlay control (K1 – K20, excluding K17 and K20). In figure 11 we present experimental results using a low transmission mask (1.4%), exposed with high exposure dose (50 mJ/cm2) at maximum wafer throughput. For this extreme use case we are able to reduce to intra-field overlay effect to below 2-nm, which is about factor 2 better than the control based on 15 parameters.

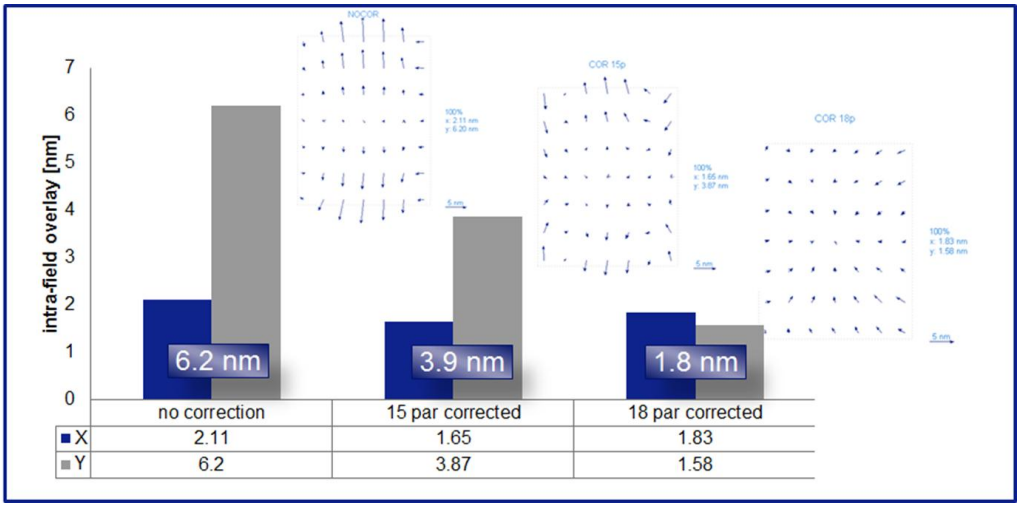

Figure 11: Experimental results of reticle heating induced intra-field overlay after applying 15 parameter and 18 parameter correction.

4.4 Experimental results: Product overlay control

In a wafer fab GridMapper [8] is used to fine tune the overlay. GridMapper is using higher order grid corrections and higher order intrafield corrections to match the expose grid to the desired application specific grid. GridMapper can also apply linear corrections per exposure (CPE). The corrections are applied per lot.

For more advanced overlay optimization we extended this functionality to enable more frequent updates and to have the high order intra-field parameters also available for corrections per exposure (HO-CPE). In the initial phase 12 parameters are available, but in the near future this will be extended to 15 parameters and 19 parameters depending on scanner system type. With such a large degree of correction freedom, the wafer metrology burden may explode, since these 19 parameters need at least 19 measurements per field, which for a full wafer with about 100 fields already is summing up to about 2000 measurements. This is not practical for high volume, and that's why HO-CPE needs fingerprint model estimation to reduce the metrology burden. In figure 12 we illustrate the overlay optimization flow for HO-CPE.

The flow starts with overlay metrology target design and target distribution over the mask. For the designed target the metrology recipe is created as well. After the reticle is fabricated, the first lot is exposed and the overlay metrology is done on the full wafer layout. For this lot, typically 4 wafers with 1000 – 2000 points per wafer are measured on the stand-alone angle resolved scatterometer. The data is send to the off-tool application server (LCP) and here the wafer fingerprint is calculated. Based on this data a reduced overlay sample scheme is derived, resulting in about 200 usable metrology points per wafer. Together with the fingerprint estimation model the sample points give sufficient information for the calculation of the HO-CPE's. The initialization is followed by the high volume production runs. Here the metrology can be done on the wafer track integrated scatterometer to enable fast feedback. Using the method of exponentially weighted moving average (EWMA) the updated scanner recipe for HO-CPE is calculated.

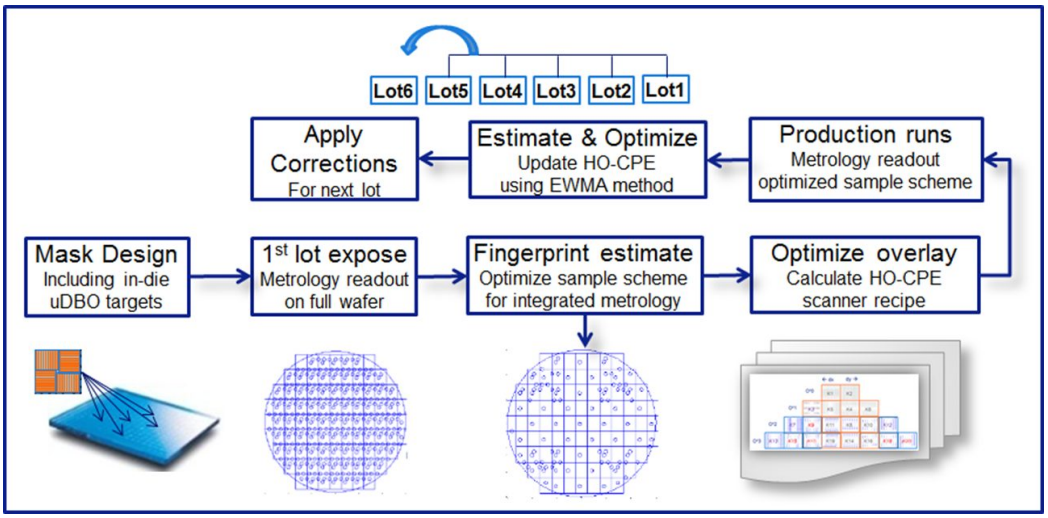

Figure 12: Overlay optimization flow from target design to control on product wafers using dynamic HO-CPE

In figures 13 and 14 we present experimental results on the product overlay improvements achieved with this method. The reported overlay numbers are based on full wafer readout (used to verify the benefit of this method). In figure 13 results are presented for a memory use case. In the process of record we see especially at the wafer edge larger overlay errors. After applying the dynamic HO-CPE method (based on 12 par) we reduce these effects and as a result the product overlay performance is reduced with about 2-nm towards 8-nm. Another use case is based on a product like test wafer, the results of which are given in figure 14. The achieved overlay with HO-CPE of 4.5-nm is about 1-nm lower than the process of record based on the linear CPE.

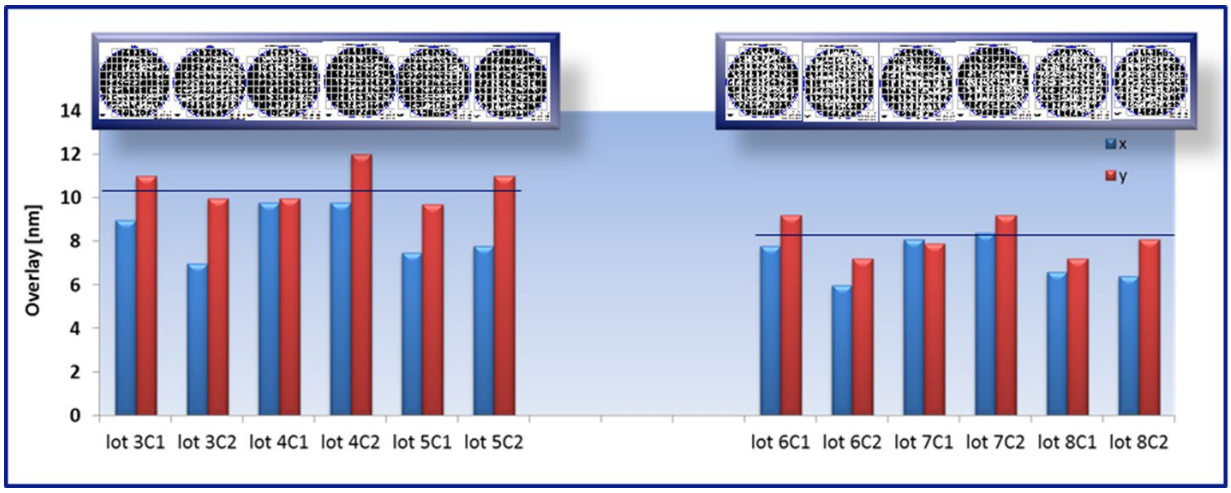

Figure 13: On product overlay results for a DRAM case on the left the process of record, and on the right the result after applying an EMWA based HO-CPE.

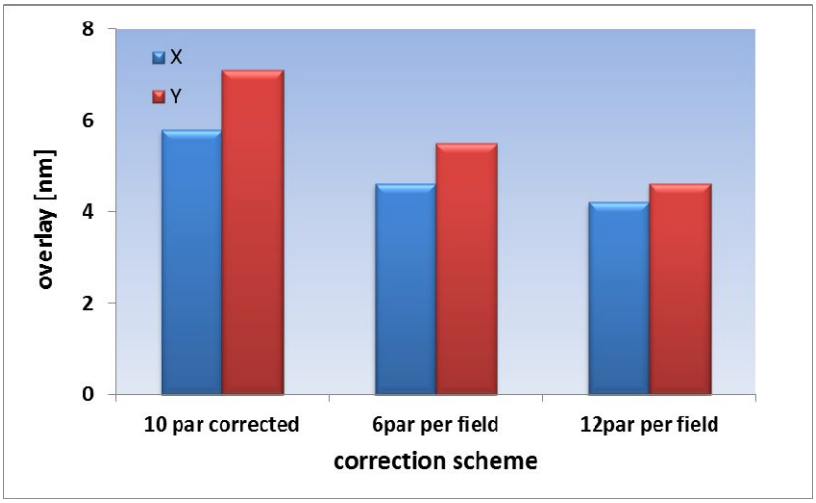

Figure 14: On product overlay results for a product like test wafer for three different optimization schemes.

5. SUMMARY AND CONCLUSIONS

Immersion lithography is being extended to the 14-nm node and as a consequence the imaging and overlay performance of the 1.35 NA scanner tools need to be improved towards 1-nm CDU and 4-nm on-product overlay. Next to generic scanner system improvements, application specific solutions are needed to follow the requirements for CD control and overlay. The application specific improvements are based on a combination of wafer based system tuning and computational optimizations of the lithography process.

In this paper we described the use of high-order field to field corrections to improve CD control, focus control and overlay control. The experimental data presented in this paper shows that with these corrections the lithography performance can be improved with 20 % to 50 %, depending on the exact case. These performance improvements enable the use of ArF immersion for the 20-nm node and 14-nm node.

6. ACKNOWLEDGMENTS

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